

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This automated official refresh report closes `polished-dataset RCIM` and `full model-development refresh` .

Decision

- `polished-dataset RCIM` and `full model-development refresh` is accepted and closed as the official `2026-07-03 TE Curve Verification Pipeline` refresh.
- The accepted polished model-development baseline is the `polished_periodic_gru_sequence` family: `polished_periodic_gru_sequence_Bw` is the strongest refreshed aggregate candidate and `polished_periodic_gru_sequence_global` is the strongest refreshed global candidate.
- The forward raw-error leader is `polished_rcim_model_bank_reproduction_ET19_Fw` . It is recorded as the polished RCIM reference-bank forward leader, while the full historical matrix forward leader remains `rcim_retuned_GBM19_Fw` and `polished_periodic_gru_sequence_Fw` remains the strongest polished model-development forward fallback.
- The accepted polished direction-parallel leaders are therefore:
`polished_rcim_model_bank_reproduction_ET19_Fw` for forward reference-bank evidence,
`polished_periodic_gru_sequence_Fw` for forward model-development evidence,
`polished_periodic_gru_sequence_Bw` for backward, and `polished_periodic_gru_sequence_global` for global.
- The launcher-generated matrix, collage, overlay, and PDF exports were reviewed during Codex closure and the status documents were synchronized.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/2026-07-03-17-46-19__track2_full_directional_family_matrix_polished_dataset_te_curve_verification_refresh_2026_07_03_fix3/validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/2026-07-03-17-46-19__track2_full_directional_family_matrix_polished_dataset_te_curve_verification_refresh_2026_07_03_fix3/per_condition_metrics.csv` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-07-03]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-07-03]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs:
`output/validation_checks/track2_operator_launch_logs/2026-07-03-18-43-42_polished_dataset_te_curve_verification_refresh_2026_07_03_fix3` .

Acceptance Checks

The accepted matrix is the `fix3` matrix generated on `2026-07-03` .

Check	Result
Dataset root	<code>data\polished_dataset</code>
Refresh source contract	<code>data\polished_dataset</code>
Historical baseline source contract	<code>data\simplified_dataset</code>
Candidate count	<code>293</code>
Refreshed candidate count	<code>128</code>
Polished refreshed per-condition rows checked	<code>15,908</code>
Polished refreshed rows with a non-polished source path	<code>0</code>

The remaining `data/simplified_dataset` reference in the matrix summary is the explicit historical baseline contract for pre-polished candidates, not the source used by the refreshed polished candidates.

Candidate Refresh

The refresh added `128` candidates from `polished_model_development_registry` , `polished_rcim_model_bank_reproduction` into the official `293` -candidate matrix.

Surface	Candidate	Family
Fw	polished_rcim_model_bank_reproduction_SVM19_Fw	SVM
Fw	polished_rcim_model_bank_reproduction_MLP19_Fw	MLP
Fw	polished_rcim_model_bank_reproduction_RF19_Fw	RF
Fw	polished_rcim_model_bank_reproduction_DT19_Fw	DT
Fw	polished_rcim_model_bank_reproduction_ET19_Fw	ET
Fw	polished_rcim_model_bank_reproduction_ERT19_Fw	ERT
Fw	polished_rcim_model_bank_reproduction_GBM19_Fw	GBM
Fw	polished_rcim_model_bank_reproduction_HGBM19_Fw	HGBM
Fw	polished_rcim_model_bank_reproduction_XGBM19_Fw	XGBM
Fw	polished_rcim_model_bank_reproduction_LGBM19_Fw	LGBM
Bw	polished_rcim_model_bank_reproduction_SVM19_Bw	SVM
Bw	polished_rcim_model_bank_reproduction_MLP19_Bw	MLP
Bw	polished_rcim_model_bank_reproduction_RF19_Bw	RF
Bw	polished_rcim_model_bank_reproduction_DT19_Bw	DT
Bw	polished_rcim_model_bank_reproduction_ET19_Bw	ET
Bw	polished_rcim_model_bank_reproduction_ERT19_Bw	ERT
Bw	polished_rcim_model_bank_reproduction_GBM19_Bw	GBM
Bw	polished_rcim_model_bank_reproduction_HGBM19_Bw	HGBM
Bw	polished_rcim_model_bank_reproduction_XGBM19_Bw	XGBM
Bw	polished_rcim_model_bank_reproduction_LGBM19_Bw	LGBM
global	polished_feedforward_global	feedforward
Fw	polished_feedforward_Fw	feedforward
Bw	polished_feedforward_Bw	feedforward
global	polished_harmonic_regression_global	harmonic_regression
Fw	polished_harmonic_regression_Fw	harmonic_regression
Bw	polished_harmonic_regression_Bw	harmonic_regression
global	polished_periodic_mlp_global	periodic_mlp

Surface	Candidate	Family
Fw	polished_periodic_mlp_Fw	periodic_mlp
Bw	polished_periodic_mlp_Bw	periodic_mlp
global	polished_residual_harmonic_mlp_global	residual_harmonic_mlp
Fw	polished_residual_harmonic_mlp_Fw	residual_harmonic_mlp
Bw	polished_residual_harmonic_mlp_Bw	residual_harmonic_mlp
global	polished_tree_global	tree
Fw	polished_tree_Fw	tree
Bw	polished_tree_Bw	tree
global	polished_periodic_mlp_harmonic_global	periodic_mlp_harmonic
Fw	polished_periodic_mlp_harmonic_Fw	periodic_mlp_harmonic
Bw	polished_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic
global	polished_temporal_convolution_global	temporal_convolution
Fw	polished_temporal_convolution_Fw	temporal_convolution
Bw	polished_temporal_convolution_Bw	temporal_convolution
global	polished_gru_sequence_global	gru_sequence
Fw	polished_gru_sequence_Fw	gru_sequence
Bw	polished_gru_sequence_Bw	gru_sequence
global	polished_lstm_sequence_global	lstm_sequence
Fw	polished_lstm_sequence_Fw	lstm_sequence
Bw	polished_lstm_sequence_Bw	lstm_sequence
global	polished_periodic_temporal_convolution_global	periodic_temporal_convolution
Fw	polished_periodic_temporal_convolution_Fw	periodic_temporal_convolution
Bw	polished_periodic_temporal_convolution_Bw	periodic_temporal_convolution
global	polished_periodic_gru_sequence_global	periodic_gru_sequence
Fw	polished_periodic_gru_sequence_Fw	periodic_gru_sequence
Bw	polished_periodic_gru_sequence_Bw	periodic_gru_sequence
global	polished_periodic_lstm_sequence_global	periodic_lstm_sequence
Fw	polished_periodic_lstm_sequence_Fw	periodic_lstm_sequence
Bw	polished_periodic_lstm_sequence_Bw	periodic_lstm_sequence

Surface	Candidate	Family
global	polished_residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim
Fw	polished_residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim
Bw	polished_residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim
global	polished_residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240
Fw	polished_residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240
Bw	polished_residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240
global	polished_residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360
Fw	polished_residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360
Bw	polished_residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360
global	polished_residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim
Fw	polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim
Bw	polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim
global	polished_residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240
Fw	polished_residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240
Bw	polished_residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240
global	polished_residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360
Fw	polished_residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360
Bw	polished_residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360
global	polished_wave3_1_sequential_residual_offset_probe_global	wave3_1_sequential_residual_offset_probe
Fw	polished_wave3_1_sequential_residual_offset_probe_Fw	wave3_1_sequential_residual_offset_probe
Bw	polished_wave3_1_sequential_residual_offset_probe_Bw	wave3_1_sequential_residual_offset_probe
global	polished_wave3_2_clean_sequential_residual_offset_global	wave3_2_clean_sequential_residual_offset
Fw	polished_wave3_2_clean_sequential_residual_offset_Fw	wave3_2_clean_sequential_residual_offset
Bw	polished_wave3_2_clean_sequential_residual_offset_Bw	wave3_2_clean_sequential_residual_offset

Surface	Candidate	Family
global	polished_wave3_2_harmonic_residual_offset_global	wave3_2_harmonic_residual_offset
Fw	polished_wave3_2_harmonic_residual_offset_Fw	wave3_2_harmonic_residual_offset
Bw	polished_wave3_2_harmonic_residual_offset_Bw	wave3_2_harmonic_residual_offset
global	polished_wave3_3_curve_aware_pointwise_control_global	wave3_3_curve_aware_pointwise_control
Fw	polished_wave3_3_curve_aware_pointwise_control_Fw	wave3_3_curve_aware_pointwise_control
Bw	polished_wave3_3_curve_aware_pointwise_control_Bw	wave3_3_curve_aware_pointwise_control
global	polished_wave3_3_raw_centered_shape_curve_aware_global	wave3_3_raw_centered_shape_curve_aware
Fw	polished_wave3_3_raw_centered_shape_curve_aware_Fw	wave3_3_raw_centered_shape_curve_aware
Bw	polished_wave3_3_raw_centered_shape_curve_aware_Bw	wave3_3_raw_centered_shape_curve_aware
global	polished_wave3_3_raw_offset_curve_aware_global	wave3_3_raw_offset_curve_aware
Fw	polished_wave3_3_raw_offset_curve_aware_Fw	wave3_3_raw_offset_curve_aware
Bw	polished_wave3_3_raw_offset_curve_aware_Bw	wave3_3_raw_offset_curve_aware
global	polished_wave3_3_full_curve_composite_global	wave3_3_full_curve_composite
Fw	polished_wave3_3_full_curve_composite_Fw	wave3_3_full_curve_composite
Bw	polished_wave3_3_full_curve_composite_Bw	wave3_3_full_curve_composite
global	polished_wave4_1_mae_robust_loss_global	wave4_1_mae_robust_loss
Fw	polished_wave4_1_mae_robust_loss_Fw	wave4_1_mae_robust_loss
Bw	polished_wave4_1_mae_robust_loss_Bw	wave4_1_mae_robust_loss
global	polished_wave4_1_smooth_l1_robust_loss_global	wave4_1_smooth_l1_robust_loss
Fw	polished_wave4_1_smooth_l1_robust_loss_Fw	wave4_1_smooth_l1_robust_loss
Bw	polished_wave4_1_smooth_l1_robust_loss_Bw	wave4_1_smooth_l1_robust_loss
global	polished_wave4_1_log_cosh_robust_loss_global	wave4_1_log_cosh_robust_loss
Fw	polished_wave4_1_log_cosh_robust_loss_Fw	wave4_1_log_cosh_robust_loss
Bw	polished_wave4_1_log_cosh_robust_loss_Bw	wave4_1_log_cosh_robust_loss
global	polished_wave4_2_quantile_p10_p50_p90_global	wave4_2_quantile_p10_p50_p90

Surface	Candidate	Family
Fw	polished_wave4_2_quantile_p10_p50_p90_Fw	wave4_2_quantile_p10_p50_p90
Bw	polished_wave4_2_quantile_p10_p50_p90_Bw	wave4_2_quantile_p10_p50_p90
global	polished_wave4_2_gaussian_nll_global	wave4_2_gaussian_nll
Fw	polished_wave4_2_gaussian_nll_Fw	wave4_2_gaussian_nll
Bw	polished_wave4_2_gaussian_nll_Bw	wave4_2_gaussian_nll
global	polished_wave4_3_mixture_density_k2_global	wave4_3_mixture_density_k2
Fw	polished_wave4_3_mixture_density_k2_Fw	wave4_3_mixture_density_k2
Bw	polished_wave4_3_mixture_density_k2_Bw	wave4_3_mixture_density_k2
global	polished_wave4_3_mixture_density_k3_global	wave4_3_mixture_density_k3
Fw	polished_wave4_3_mixture_density_k3_Fw	wave4_3_mixture_density_k3
Bw	polished_wave4_3_mixture_density_k3_Bw	wave4_3_mixture_density_k3
global	polished_wave4_4_gru_latent_offset_residual_global	wave4_4_gru_latent_offset_residual
Fw	polished_wave4_4_gru_latent_offset_residual_Fw	wave4_4_gru_latent_offset_residual
Bw	polished_wave4_4_gru_latent_offset_residual_Bw	wave4_4_gru_latent_offset_residual
global	polished_wave4_4_causal_tcn_latent_offset_residual_global	wave4_4_causal_tcn_latent_offset_residual
Fw	polished_wave4_4_causal_tcn_latent_offset_residual_Fw	wave4_4_causal_tcn_latent_offset_residual
Bw	polished_wave4_4_causal_tcn_latent_offset_residual_Bw	wave4_4_causal_tcn_latent_offset_residual
global	polished_wave5_1_harmonic_prior_pointwise_control_global	wave5_1_harmonic_prior_pointwise_control
Fw	polished_wave5_1_harmonic_prior_pointwise_control_Fw	wave5_1_harmonic_prior_pointwise_control
Bw	polished_wave5_1_harmonic_prior_pointwise_control_Bw	wave5_1_harmonic_prior_pointwise_control
global	polished_wave5_1_harmonic_prior_smooth_l1_structured_global	wave5_1_harmonic_prior_smooth_l1_structured
Fw	polished_wave5_1_harmonic_prior_smooth_l1_structured_Fw	wave5_1_harmonic_prior_smooth_l1_structured
Bw	polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw	wave5_1_harmonic_prior_smooth_l1_structured

Refreshed Source Leaders

The table ranks the refreshed source by aggregate offline TE Curve Verification Pipeline metrics.

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	polished_rcim_model_bank_reproduction_ET19_Fw	0.001155	0.001394	2.401	4.928
Bw	polished_periodic_gru_sequence_Bw	0.001129	0.001412	2.228	4.688
global	polished_periodic_gru_sequence_global	0.001279	0.001568	2.636	5.690

Refreshed Source Leaderboard

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
1	Bw	polished_periodic_gru_sequence_Bw	0.001129	0.001412	2.228	4.688
2	Fw	polished_rcim_model_bank_reproduction_ET19_Fw	0.001155	0.001394	2.401	4.928
3	Fw	polished_rcim_model_bank_reproduction_ERT19_Fw	0.001187	0.001438	2.457	5.438
4	Fw	polished_periodic_gru_sequence_Fw	0.001195	0.001461	2.559	5.193
5	global	polished_periodic_gru_sequence_global	0.001279	0.001568	2.636	5.690
6	Bw	polished_periodic_lstm_sequence_Bw	0.001290	0.001613	2.539	4.524
7	global	polished_periodic_lstm_sequence_global	0.001303	0.001600	2.642	5.314
8	Fw	polished_rcim_model_bank_reproduction_LGBM19_Fw	0.001368	0.001635	2.864	6.832
9	Fw	polished_wave4_3_mixture_density_k3_Fw	0.001528	0.001867	3.161	7.018
10	Fw	polished_wave4_3_mixture_density_k2_Fw	0.001545	0.001890	3.202	7.033
11	global	polished_wave4_3_mixture_density_k3_global	0.001585	0.001945	3.116	7.127
12	Fw	polished_rcim_model_bank_reproduction_GBM19_Fw	0.001644	0.001903	3.449	9.961
13	Fw	polished_wave3_3_curve_aware_pointwise_control_Fw	0.001701	0.002055	3.407	7.927
14	Fw	polished_wave4_2_gaussian_nll_Fw	0.001711	0.002056	3.424	8.122
15	Fw	polished_wave3_3_raw_centered_shape_curve_aware_Fw	0.001716	0.002078	3.450	8.345
16	Fw	polished_wave4_2_quantile_p10_p50_p90_Fw	0.001727	0.002081	3.466	7.927

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
17	Fw	polished_periodic_lstm_sequence_Fw	0.001730	0.002084	3.517	7.494
18	Fw	polished_wave3_3_raw_offset_curve_aware_Fw	0.001734	0.002099	3.495	8.291
19	Fw	polished_periodic_mlp_harmonic_Fw	0.001735	0.002062	3.511	7.554
20	Fw	polished_wave3_2_harmonic_residual_offset_Fw	0.001756	0.002127	3.526	8.417
21	global	polished_wave4_3_mixture_density_k2_global	0.001758	0.002131	3.354	8.461
22	Fw	polished_wave4_1_log_cosh_robust_loss_Fw	0.001764	0.002125	3.566	8.171
23	Fw	polished_wave4_1_mae_robust_loss_Fw	0.001775	0.002132	3.569	8.373
24	Fw	polished_wave3_3_full_curve_composite_Fw	0.001786	0.002167	3.606	8.311
25	Fw	polished_wave5_1_harmonic_prior_smooth_l1_structured_Fw	0.001795	0.002160	3.630	8.304
26	Fw	polished_rcim_model_bank_reproduction_SVM19_Fw	0.001797	0.002145	3.616	10.200
27	Fw	polished_wave4_1_smooth_l1_robust_loss_Fw	0.001799	0.002176	3.645	8.067
28	Fw	polished_residual_harmonic_gru_sequence_sparse_rcim_Fw	0.001832	0.002216	3.700	8.239
29	Fw	polished_wave5_1_harmonic_prior_pointwise_control_Fw	0.001881	0.002253	3.835	8.184
30	Fw	polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.001892	0.002286	3.839	8.161
31	global	polished_wave4_1_mae_robust_loss_global	0.001907	0.002311	3.588	9.258
32	global	polished_wave4_2_quantile_p10_p50_p90_global	0.001908	0.002311	3.589	9.208
33	global	polished_wave4_1_log_cosh_robust_loss_global	0.001923	0.002317	3.610	9.289
34	Bw	polished_wave4_3_mixture_density_k3_Bw	0.001930	0.002341	3.405	9.512
35	global	polished_wave3_2_harmonic_residual_offset_global	0.001936	0.002349	3.656	9.145
36	global	polished_wave3_3_curve_aware_pointwise_control_global	0.001967	0.002372	3.719	9.186
37	global	polished_wave3_3_raw_centered_shape_curve_aware_global	0.001977	0.002372	3.710	9.358
38	Bw	polished_wave4_3_mixture_density_k2_Bw	0.001995	0.002403	3.526	9.548
39	Fw	polished_rcim_model_bank_reproduction_RF19_Fw	0.001998	0.002293	4.243	10.816
40	Fw	polished_periodic_temporal_convolution_Fw	0.002004	0.002385	4.123	9.059
41	global	polished_wave4_2_gaussian_nll_global	0.002006	0.002417	3.776	9.403
42	global	polished_wave3_3_raw_offset_curve_aware_global	0.002009	0.002428	3.824	9.323

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
43	global	polished_wave4_1_smooth_l1_robust_loss_global	0.002018	0.002417	3.821	9.619
44	global	polished_wave3_3_full_curve_composite_global	0.002028	0.002464	3.869	9.205
45	Fw	polished_wave3_2_clean_sequential_residual_offset_Fw	0.002052	0.002494	4.192	8.319
46	global	polished_periodic_mlp_harmonic_global	0.002054	0.002457	3.832	9.502
47	global	polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.002069	0.002515	3.949	9.574
48	Fw	polished_wave3_1_sequential_residual_offset_probe_Fw	0.002071	0.002524	4.236	8.319
49	Fw	polished_lstm_sequence_Fw	0.002083	0.002538	4.261	8.262
50	Fw	polished_residual_harmonic_mlp_Fw	0.002093	0.002541	4.287	8.352
51	Fw	polished_gru_sequence_Fw	0.002102	0.002552	4.308	8.229
52	global	polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.002107	0.002548	4.033	9.587
53	Fw	polished_periodic_mlp_Fw	0.002118	0.002575	4.351	8.283
54	global	polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.002121	0.002522	3.936	9.529
55	Fw	polished_tree_Fw	0.002125	0.002612	4.355	8.534
56	Fw	polished_feedforward_Fw	0.002130	0.002586	4.378	8.431
57	Bw	polished_wave4_1_log_cosh_robust_loss_Bw	0.002131	0.002576	3.787	9.666
58	Bw	polished_wave4_1_mae_robust_loss_Bw	0.002133	0.002572	3.754	9.697
59	Bw	polished_wave4_2_gaussian_nll_Bw	0.002133	0.002582	3.758	9.788
60	Bw	polished_wave3_3_raw_centered_shape_curve_aware_Bw	0.002133	0.002578	3.790	9.660
61	Bw	polished_wave4_2_quantile_p10_p50_p90_Bw	0.002133	0.002585	3.778	9.700
62	Fw	polished_wave4_4_gru_latent_offset_residual_Fw	0.002135	0.002611	4.373	8.386
63	Bw	polished_wave3_3_raw_offset_curve_aware_Bw	0.002139	0.002591	3.806	9.694
64	Bw	polished_wave3_2_harmonic_residual_offset_Bw	0.002142	0.002591	3.805	9.658
65	Fw	polished_wave4_4_causal_tcn_latent_offset_residual_Fw	0.002149	0.002620	4.408	8.453
66	global	polished_wave5_1_harmonic_prior_pointwise_control_global	0.002153	0.002556	4.001	9.336
67	Bw	polished_wave3_3_curve_aware_pointwise_control_Bw	0.002172	0.002638	3.909	9.815
68	Fw	polished_temporal_convolution_Fw	0.002210	0.002686	4.557	8.559
69	global	polished_gru_sequence_global	0.002235	0.002722	4.313	9.564

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
70	Bw	polished_wave4_1_smooth_l1_robust_loss_Bw	0.002236	0.002696	4.026	9.581
71	global	polished_wave3_1_sequential_residual_offset_probe_global	0.002252	0.002737	4.348	9.622
72	global	polished_lstm_sequence_global	0.002266	0.002753	4.390	9.554
73	global	polished_wave3_2_clean_sequential_residual_offset_global	0.002276	0.002760	4.397	9.579
74	global	polished_wave4_4_gru_latent_offset_residual_global	0.002306	0.002807	4.470	9.612
75	global	polished_periodic_temporal_convolution_global	0.002318	0.002743	4.527	9.986
76	global	polished_wave4_4_causal_tcn_latent_offset_residual_global	0.002321	0.002831	4.493	9.907
77	Bw	polished_periodic_temporal_convolution_Bw	0.002326	0.002803	4.277	9.862
78	Bw	polished_residual_harmonic_gru_sequence_sparse_rcim_Bw	0.002331	0.002829	4.234	9.948
79	Bw	polished_wave3_3_full_curve_composite_Bw	0.002333	0.002822	4.250	9.763
80	Bw	polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.002343	0.002825	4.242	9.919
81	Fw	polished_rcim_model_bank_reproduction_DT19_Fw	0.002373	0.002664	5.004	11.005
82	Bw	polished_periodic_mlp_harmonic_Bw	0.002396	0.002823	4.137	10.607
83	global	polished_temporal_convolution_global	0.002398	0.002909	4.702	9.713
84	global	polished_feedforward_global	0.002404	0.002907	4.575	10.056
85	Bw	polished_wave3_1_sequential_residual_offset_probe_Bw	0.002411	0.002947	4.412	10.138
86	global	polished_residual_harmonic_mlp_global	0.002411	0.002910	4.592	10.130
87	Bw	polished_wave5_1_harmonic_prior_pointwise_control_Bw	0.002418	0.002843	4.202	10.476
88	global	polished_periodic_mlp_global	0.002421	0.002922	4.606	9.853
89	Bw	polished_gru_sequence_Bw	0.002425	0.002937	4.438	10.148
90	Bw	polished_lstm_sequence_Bw	0.002430	0.002973	4.452	10.190
91	global	polished_tree_global	0.002431	0.002939	4.635	9.931
92	Bw	polished_wave3_2_clean_sequential_residual_offset_Bw	0.002439	0.002959	4.469	10.192
93	Bw	polished_wave4_4_gru_latent_offset_residual_Bw	0.002455	0.002998	4.512	10.190
94	Bw	polished_wave4_4_causal_tcn_latent_offset_residual_Bw	0.002485	0.003022	4.545	10.210
95	Bw	polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw	0.002528	0.002976	4.377	10.763
96	Bw	polished_temporal_convolution_Bw	0.002530	0.003060	4.674	10.195
97	Fw	polished_rcim_model_bank_reproduction_HGBM19_Fw	0.002606	0.002910	5.628	12.844

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
98	Bw	polished_feedforward_Bw	0.002655	0.003193	4.708	10.602
99	Bw	polished_residual_harmonic_mlp_Bw	0.002713	0.003255	4.822	10.624
100	Bw	polished_tree_Bw	0.002756	0.003287	4.934	10.752
101	Bw	polished_periodic_mlp_Bw	0.002769	0.003282	4.910	10.607
102	Fw	polished_rcim_model_bank_reproduction_XGBM19_Fw	0.003028	0.003304	6.606	18.657
103	global	polished_residual_harmonic_gru_sequence_dense240_global	0.003121	0.003993	6.395	10.129
104	Fw	polished_residual_harmonic_gru_sequence_dense240_Fw	0.003186	0.004142	6.811	9.157
105	Fw	polished_residual_harmonic_lstm_sequence_dense240_Fw	0.003240	0.004196	6.948	9.142
106	global	polished_residual_harmonic_lstm_sequence_dense240_global	0.003329	0.004300	6.862	10.535
107	Bw	polished_residual_harmonic_gru_sequence_dense240_Bw	0.003416	0.004405	6.794	10.465
108	Bw	polished_residual_harmonic_lstm_sequence_dense240_Bw	0.003569	0.004639	7.137	10.677
109	Bw	polished_rcim_model_bank_reproduction_ERT19_Bw	0.003715	0.003957	7.516	43.149
110	Bw	polished_rcim_model_bank_reproduction_LGBM19_Bw	0.003765	0.004028	7.677	43.640
111	global	polished_harmonic_regression_global	0.003979	0.004503	8.163	14.763
112	Bw	polished_rcim_model_bank_reproduction_ET19_Bw	0.004034	0.004316	8.163	41.965
113	Bw	polished_rcim_model_bank_reproduction_DT19_Bw	0.004395	0.004659	9.372	39.764
114	Bw	polished_rcim_model_bank_reproduction_SVM19_Bw	0.004409	0.004851	8.793	38.822
115	Bw	polished_rcim_model_bank_reproduction_GBM19_Bw	0.004482	0.004755	9.169	43.941
116	Fw	polished_residual_harmonic_gru_sequence_dense360_Fw	0.004563	0.007628	9.896	11.402
117	global	polished_residual_harmonic_lstm_sequence_dense360_global	0.004618	0.006711	9.753	12.396
118	global	polished_residual_harmonic_gru_sequence_dense360_global	0.004631	0.006987	9.827	12.042
119	Fw	polished_residual_harmonic_lstm_sequence_dense360_Fw	0.004652	0.007237	10.094	11.605
120	Bw	polished_rcim_model_bank_reproduction_RF19_Bw	0.004852	0.005155	10.113	46.318
121	Bw	polished_rcim_model_bank_reproduction_HGBM19_Bw	0.004919	0.005254	10.239	44.526
122	Bw	polished_residual_harmonic_lstm_sequence_dense360_Bw	0.005029	0.007977	10.455	12.921
123	Bw	polished_residual_harmonic_gru_sequence_dense360_Bw	0.005031	0.008128	10.446	12.856
124	Bw	polished_rcim_model_bank_reproduction_XGBM19_Bw	0.005286	0.005591	11.237	48.248

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
125	Bw	polished_harmonic_regression_Bw	0.008041	0.008675	16.236	27.309
126	Fw	polished_rcim_model_bank_reproduction_MLP19_Fw	0.036479	0.044676	82.877	244.026
127	Fw	polished_harmonic_regression_Fw	0.062598	0.062702	133.783	271.628
128	Bw	polished_rcim_model_bank_reproduction_MLP19_Bw	0.062780	0.077368	139.565	338.222

Current Direction Leaders

These leaders are read from the full matrix direction breakdown after the refresh. This table includes historical comparison anchors as well as the new polished candidates; therefore the forward all-matrix leader can differ from the refreshed polished-source leader.

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	polished_periodic_gru_sequence_Bw	0.001129	0.001412	2.228	4.688
forward	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912

Visual Evidence

The same launcher run regenerated the visual companion reports and verified that the refreshed source appears in the visual package.

Source	Collage	Overlay Forward	Overlay Backward
polished_model_development_registry , polished_rcim_model_bank_reproduction	108	36	36

Closeout Decision

polished-dataset RCIM and full model-development refresh is closed as: accepted official TE Curve Verification Pipeline refresh on 2026-07-03 .

The polished refresh changes the accepted model-development baseline to the polished_periodic_gru_sequence family. Direction-specific evidence remains parallel rather than destructive: the historical full-matrix forward leader remains rcim_retuned_GBM19_Fw , the polished RCIM model bank supplies the best forward refreshed-source reference-bank curve candidate, and the polished periodic GRU sequence family supplies the strongest backward, global, and deployable model-development evidence.

No paper original or paper-retuned candidate was retrained by this refresh. The pre-polished historical candidates remain in the matrix only as comparison anchors.